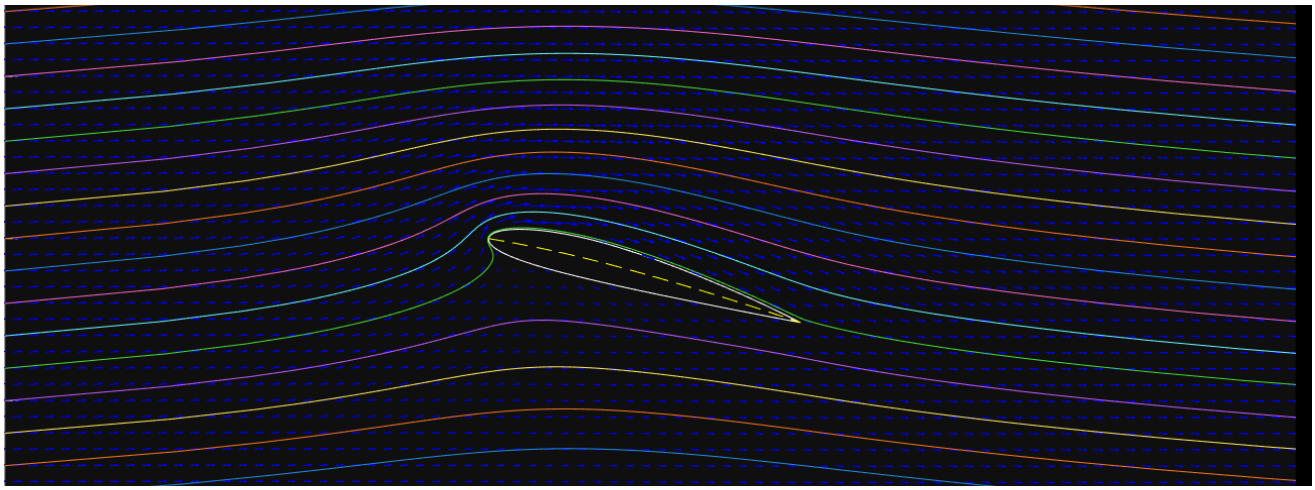
Preliminary Evaluation of Vortex Panel Method and Convergence of Numerical Solution

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Investigation Topics

1. Convergence of the Vortex Panel Method based on the number of panels and type of spacing used.
2. Differences in methods for calculating lift coefficients in Vortex Panel Method.
3. TODO: Comparison of TAT, VPM, and XFoil lift predictions.
    a. Effects of thickness on CL predictions TAT, VPM, and XFoil.
    b. Effects of camber on CL predictions on TAT, VPM, and XFoil.
    c. Difference in pressure distributions over the aerofoil.

I shall only consider NACA 4-digit aerofoils. The 1st digit represents the maximum camber from the mean chord line divided by 100c, where c is the chord length. The 2nd digit represents the position of maximum camber along the chord line divided by 10c. The last 2 digits represent the maximum thickness divided by 100c. For simplicity, a flat plate will be approximated by NACA 0001. Since the Vortex Panel Method does not model viscous effects such as the boundary layer or separation, all analysis is confined to at most 15° of angle of attack (AoA).



Formulation

Assumptions of Vortex Panel Method (VPM):

1. Steady-state
2. Inviscid fluid
3. Irrotational flow (vorticity is zero except along the aerofoil's boundaries)
4. Incompressible

Vortex Panel Method replaces the aerofoil boundaries with a vortex sheet of initially unknown vortex sheet strength. A vortex sheet is, in 2D, a line element with singular vorticity, much like a point vortex is a point singularity with vorticity, and the vortex sheet strength is the circulation around a unit length of vortex sheet. This allows circulation around the aerofoil, which, by the Kutta-Joukowski Theorem, is proportional to lift. The flow around the aerofoil must satisfy 2 conditions: no penetration through the aerofoil's boundaries and the Kutta Condition, that the flow leaves the trailing edge smoothly. The Kutta Condition also implies that the trailing edge vortex sheet strength (circulation per unit length) is 0. Through these conditions, the vortex sheet strength can be solved and therefore the velocity field around the aerofoil calculated. Further calculations yield the lift and moment predictions.

To simplify the method for numerical computation, the aerofoil's boundary is split into N discrete pieces known as panels. N is the number of panels. Each panel represents a vortex sheet of linearly varying vortex sheet strength between the vortex sheet strengths at the 2 ends. By determining the vortex sheet strengths at these 2 ends, known as the boundary vortex sheet strengths, the vortex sheet strength across the panel is fully determined. For N panels, there are N+1 boundary vortex sheet strengths to be solved (the trailing edge is double counted to allow for the application of the Kutta Condition). At each of the panel centres, we enforce the no-penetration conditions, forming N equations. Lastly, the Kutta Condition is applied at the trailing edge.

The velocity field can be calculated by a potential flow approach which, combined with Bernuolli's equation, allows determination of the pressure field. This pressure field can then be used to calculate the lift over the aerofoil at a small sample distance outside the aerofoil's boundaries. An alternative approach which will be compared is to calculate the total circulation around the aerofoil and to apply the Kutta-Joukowski Theorem ($L = \rho u \Gamma$).

Convergence of the Vortex Panel Method based on the number of panels and type of spacing used.

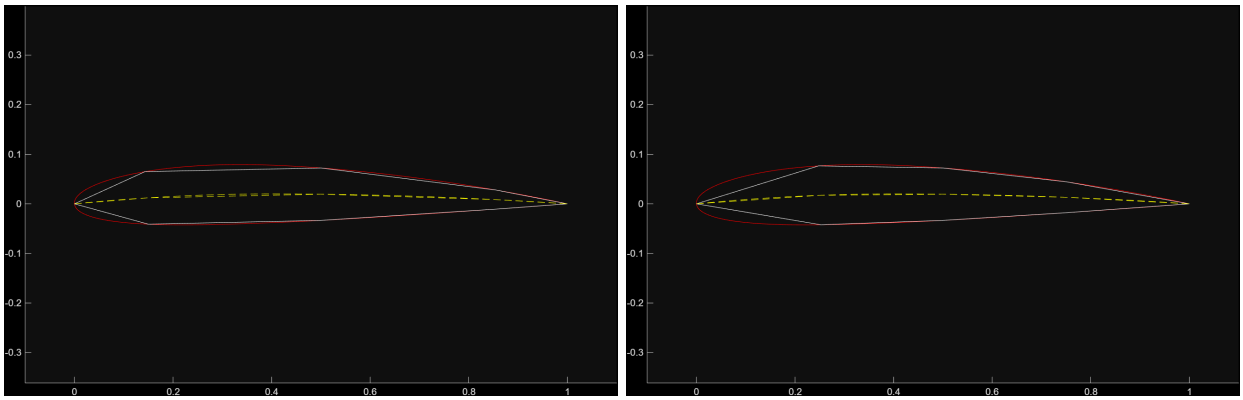
In this section, I analyse the differences between using a cosine and regular distribution. The comparison is made based on the convergence rate toward an asymptote as the number of panels is increased for both distributions. The next point is an investigation of an upper bound limit to the number of panels. Note that accuracy cannot be determined without empirical data and will therefore not be considered in this report. For visualisation, NACA 2412 will be used.

Comparison of Panel Distribution

A simple method of generating panel coordinates is to use a regular spacing along the chord line. However, the aerofoil has a 'significant' geometric feature at the leading edge due to its high curvature compared to the rest of the aerofoil. The velocity field in real flow also has 2 interesting features that can be modelled in the Vortex Panel Method close to the leading edge. Firstly, we expect a suction peak above the leading edge and the fore stagnation point below the leading edge. It is therefore sensible to increase the fidelity around the leading edge to better model the abrupt changes in the flow field near that region.

This is done by using a cosine distribution. Suppose the x-axis is aligned with the chord line. By treating $x/c = (1 - \cos\theta)/2$ for $0 < \theta < \pi/2$, and plotting panel boundaries at constant increments of θ instead of x, panels are smaller and more clumped together at the leading and trailing edges. On the contrary, a regular distribution plots panel boundaries at constant increments in x.

Images plotting a NACA 2412 for 8 panels:

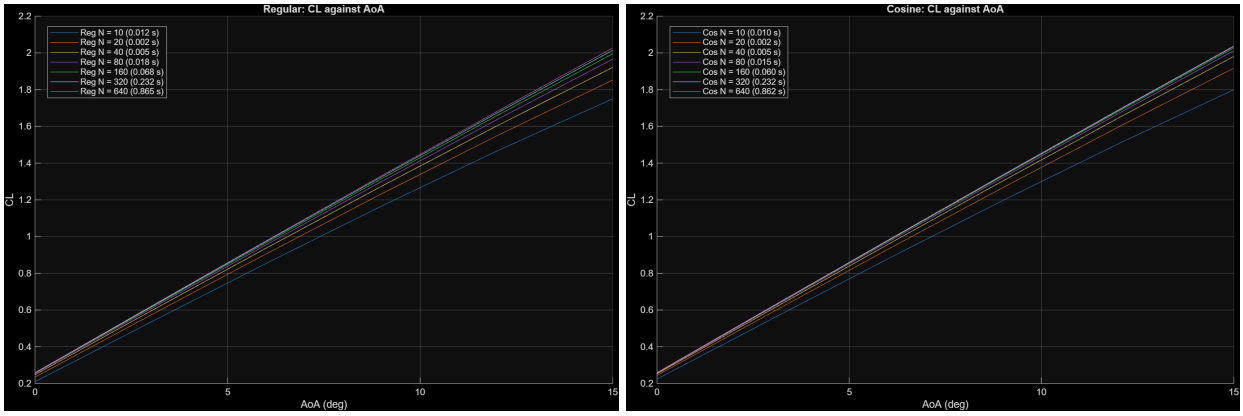


The left image uses a cosine distribution while the right uses a regular distribution. The cosine distribution better approximates the curvature around the leading edge, while the regular distribution better approximates the mid-section.

Panel Convergence Rate

To compare whether a cosine distribution is more 'economical' by focusing panels near regions of significant activity, we can study the convergence rates of the 2 distributions as the number of panels N increases.

CL-AoA curves for NACA 2412 with varying N for regular (left) and cosine (right) distributions.



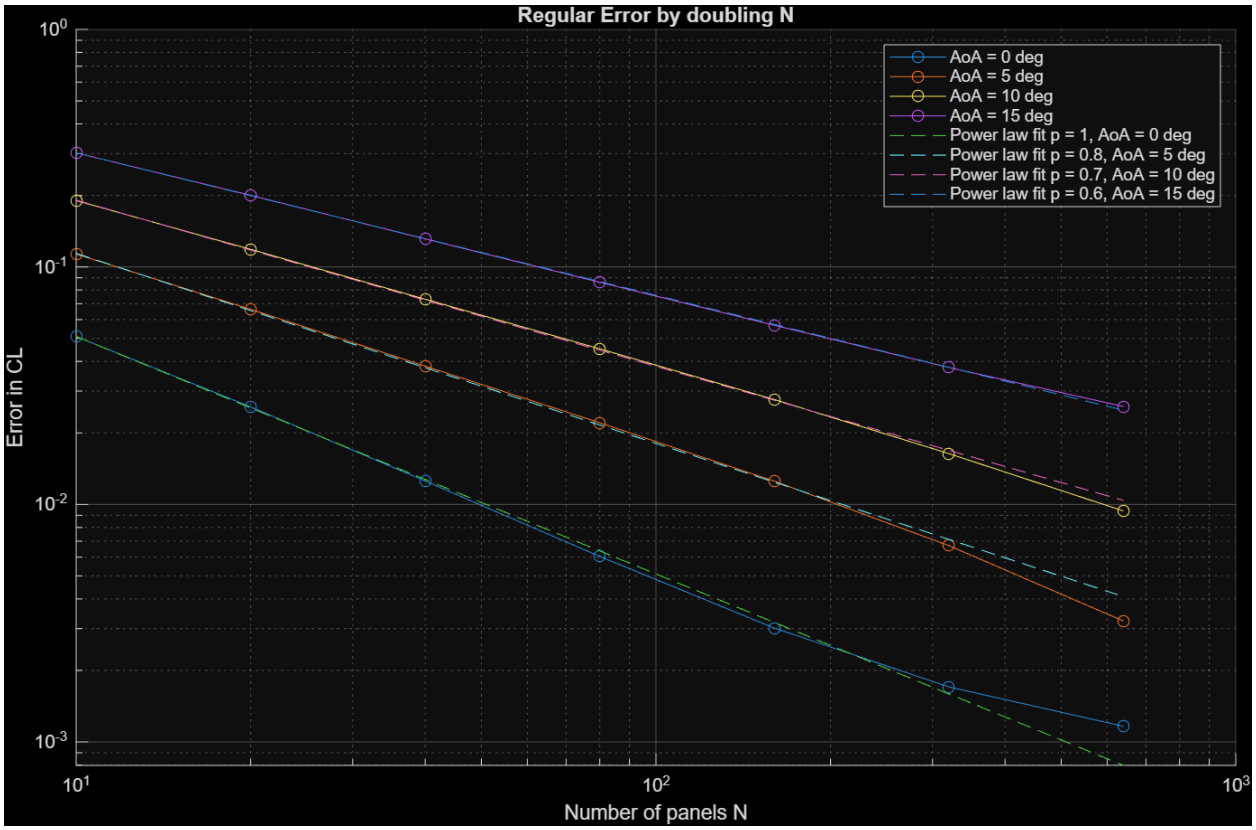
N was plotted for N = [10 20 40 80 160 320 640]. As N increases, CL increases for a given AoA. However, as N doubles, the difference in CL decreases.

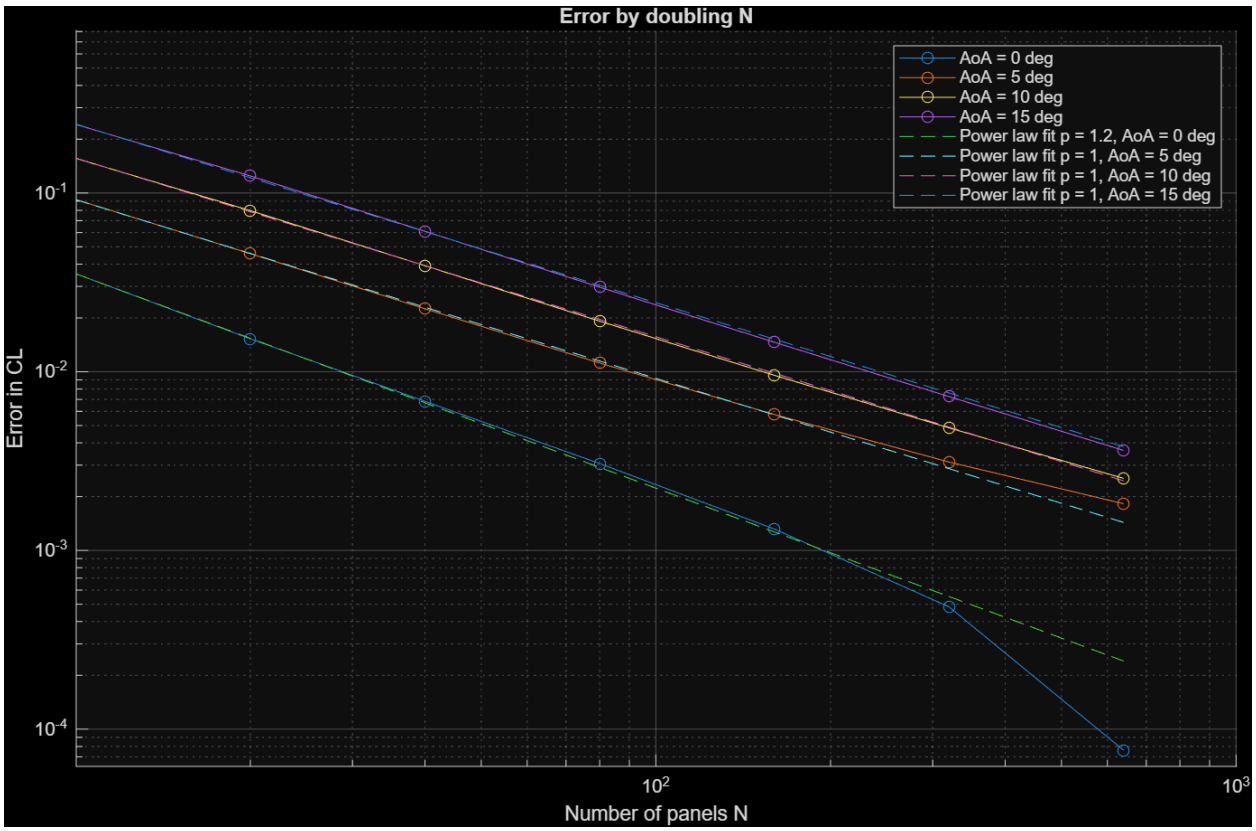
A power law of $C(N) = C(\infty) + k / N^p$ was trialled for the convergence of CL against N. $C(\infty)$ represents the asymptotic value, k is a constant, and p is an exponent that controls the rate at which the error term (k / N^p) vanishes. A greater p implies a quicker error decay.

A Richardson extrapolation was used to estimate the exponent p. Then, trial values near the estimate of p were used in a linear regression model. For each angle of attack, the exponent p that maximised the Coefficient of Determination (the best fit) was used to determine $C(\infty)$.

Using Richardson extrapolation on CL against N to estimate the exponent p showed that, for NACA 2412 with a cosine distribution, the error was strongly proportional to 1/N at non-zero AoA, while a regular distribution had an error that decreased with $p = 0.7$. The Pearson Regression Coefficient squared was greater than 99% for all fits.

$\log(CL - CL(\infty))$ against $\log(N)$ for varying AoA for regular and cosine distributions.


Regular Error by doubling N
AoA = 0 deg
AoA = 5 deg
AoA = 10 deg
AoA = 15 deg
Power law fit p = 1, AoA = 0 deg
Power law fit p = 0.8, AoA = 5 deg
Power law fit p = 0.7, AoA = 10 deg
Power law fit p = 0.6, AoA = 15 deg
Error in CL
Number of panels N


Error by doubling N
AoA = 0 deg
AoA = 5 deg
AoA = 10 deg
AoA = 15 deg
Power law fit p = 1.2, AoA = 0 deg
Power law fit p = 1, AoA = 5 deg
Power law fit p = 1, AoA = 10 deg
Power law fit p = 1, AoA = 15 deg
Error in CL
Number of panels N

For most NACA 4-digit aerofoils, the cosine distribution's error has an exponent p(Cosine) greater than the regular distribution's exponent p(Regular) by roughly 0.4. p(Cosine) - p(Regular) = 0.4. The absolute error in CL to the asymptotic values for a cosine distribution is also slightly less compared to the regular distribution. For example, at N=10, CL differs by ~0.6 for a NACA 2412 at AoA = 15°, and the error diminishes more quickly for the cosine distribution.

The power fit law also predicts an asymptotic value of CL, and these are tabulated for the NACA 2412.

Angles of Attack (°)	Asymptotic CL for regular distribution	Asymptotic CL for cosine distribution
0	0.2593	0.2604
5	0.8622	0.8610
10	1.4574	1.4582
15	2.0415	2.0524

The asymptotic values for the 2 distributions are not the same, but the variations are no greater than 1% from each other and may be considered negligible. It is highly expected that the discrepancy between the 2 will be negligible compared to other sources of error.

The table below compares exponents p for regular and cosine distributions for different aerofoils, varying in maximum camber, camber position, and thickness.

Maximum camber:

Airfoil's NACA code	Exponent p for regular distribution	Exponent p for cosine distribution
0012	0.5	0.9

2412	0.6	1
4412	0.7	1
6412	0.8	1
8412	0.8	1

Camber position:

Airfoil's NACA code	Exponent p for regular distribution	Exponent p for cosine distribution
2112	0.6	1
2312	0.6	1
2512	0.6	1
2712	0.6	1

Maximum thickness:

Airfoil's NACA code	Exponent p for regular distribution	Exponent p for cosine distribution
2410	0.3	0.9
2430	0.7	1.1
2450	0.8	1.1
2470	0.9	1.1

It is found that for aerofoils with greater camber and thickness, the convergence rate of a regular distribution improves. The greater camber and thickness could make the position of maximum camber and maximum thickness, both near the centres, of greater significance compared to smaller camber and thickness. However, the cosine distribution still outperforms the regular distribution.

Therefore, a cosine distribution, with the same number of panels, reduces the discretisation error of the method. It is found that for N = 50, the difference in CL prediction with asymptotic $CL(\infty)$ is ~ 0.05 N for N = 200, the difference reduces to 0.01. As the time of computation is practically instant (< 10 seconds), N = 200 will be used for future computation.

‘Overpacking’ Panels - an upper bound

An attempt was made to compute NACA 0001 with N = 1500 and 2000 but was unsuccessful.

A likely reason is the close spacing between panels, particularly due to the small thickness between the upper and lower surfaces and at the leading and trailing edges, where the cosine distribution of panels clumps panels closer together.

The (N+1) boundary vortex sheet strengths are solved as a system of linear equations by (N) ‘no-penetration’ conditions enforced at each panel centre and the Kutta condition at the trailing edge. No-penetration conditions are solved by summing the freestream and velocity influenced by vortex sheets and setting their normal component of velocity to be 0. The influence of a vortex sheet on another panel is heavily determined by the geometry between the two. When panels are spaced close together, adjacent vortex sheets have very similar influences on other panels, which makes the system of linear equations close to being linearly dependent. In technical terms, the influence matrix has a high condition number and is nearly singular. This makes the solution highly sensitive to small perturbations/errors in the influence coefficients forming the linear equations. Errors such as floating-point error now become non-negligible and cause unreliable solutions or none at all.

For the NACA 0001 aerofoil, the highest number of panels achieved with a cosine spacing was 1012 panels, with the smallest panel of length 9.638e-6 and a condition number of 5.031e+7.

For thicker panels like the NACA 2412, this problem only arises around N = 4500.

Differences in methods for calculating lift coefficients in Vortex Panel Method

Introduction of 3 different methods

The principal method for calculating lift is to utilise the Kutta-Joukowski Theorem: $L = \rho u \Gamma$ by calculating the total circulation around the aerofoil. This should be the most precise method, as the vortex sheet strengths are calculated and known for the aerofoil at an early stage in the Vortex Panel Method, and the circulation is simply the integral of vortex sheet strength over the entire aerofoil. Since it was defined that the vortex sheet strength across each panel varies linearly between 2 boundary values, the integral simplifies exactly to a trapezoidal rule summation. This method makes the least assumptions and simplifications to the following alternatives.

The first alternative method is to calculate pressure at sample points at a small distance ε outside the aerofoil's panel centres. We cannot sample points along the panels, as the vortex sheets are line singularities, so velocities and pressures are undefined (div by 0 error). By a classical calculus argument, for a high number of panels, each panel length becomes sufficiently small that the pressure variation across a panel is linear. If this assumption holds, the force on each panel can be simply calculated as the pressure at the centre of the panel multiplied by the panel's length, in effect a midpoint estimation of the integral of pressure across the panel. It is expected that this method of approximation will be most vulnerable at the suction peak and fore stagnation point where pressure changes are abrupt compared to the rest of the aerofoil. I will compare the pressure plots over the aerofoil's surfaces for different ε and the CL predictions with other methods.

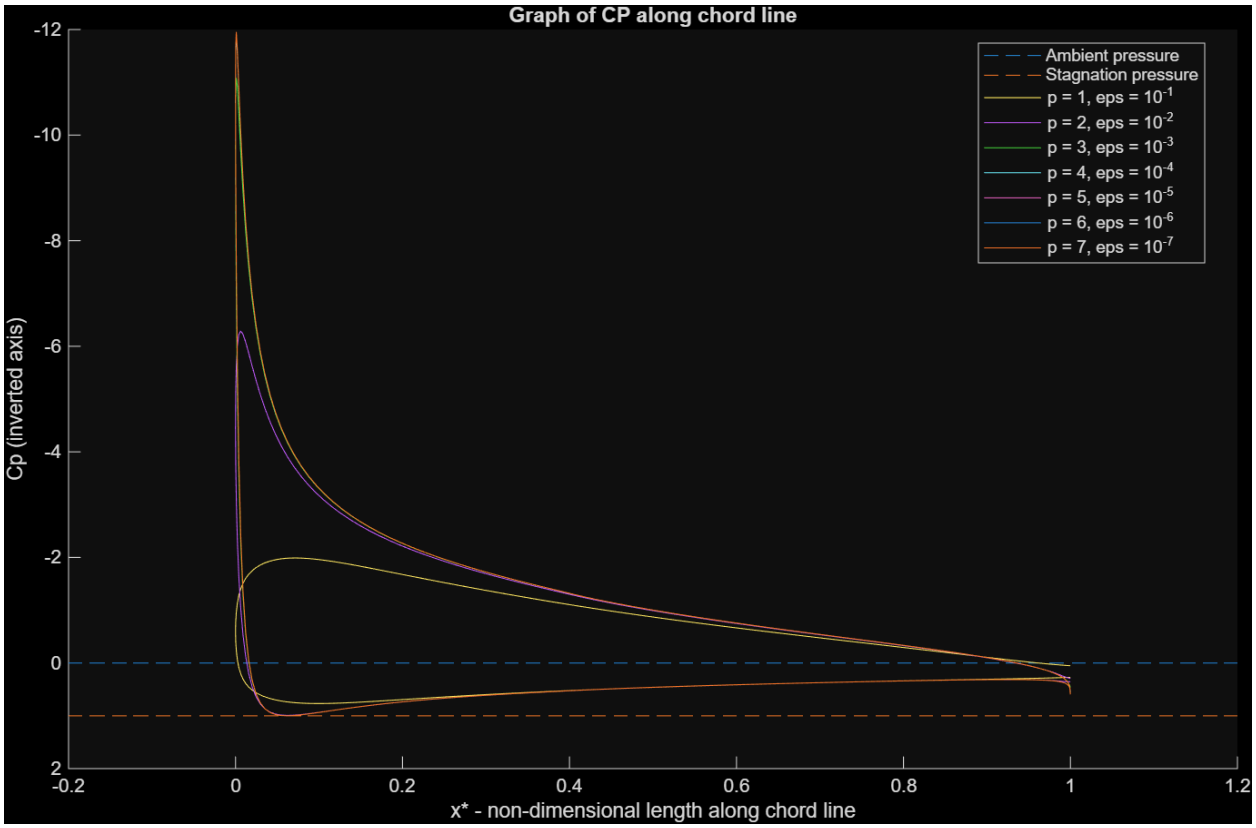
The second alternative is to compute the steady-state momentum equation on a control volume containing the aerofoil. This is simply Body Forces + Surface Forces = Momentum Flux. The only body force is the negative of lift from the aerofoil, forcing the air downwards. Surface forces are the pressure forces on the boundaries of the control volume, which can be determined

by Bernuolli's equation after determining the velocity field. Lastly, the momentum flux can be determined also from the velocity field.

Pressure Field from 1st Alternative

The main benefit of the first method is that in the process, the pressure over the airfoil can be plotted. However, the distance of the sample points away from panel centres, ε, must be appropriately small so that errors in pressure predictions are negligible.

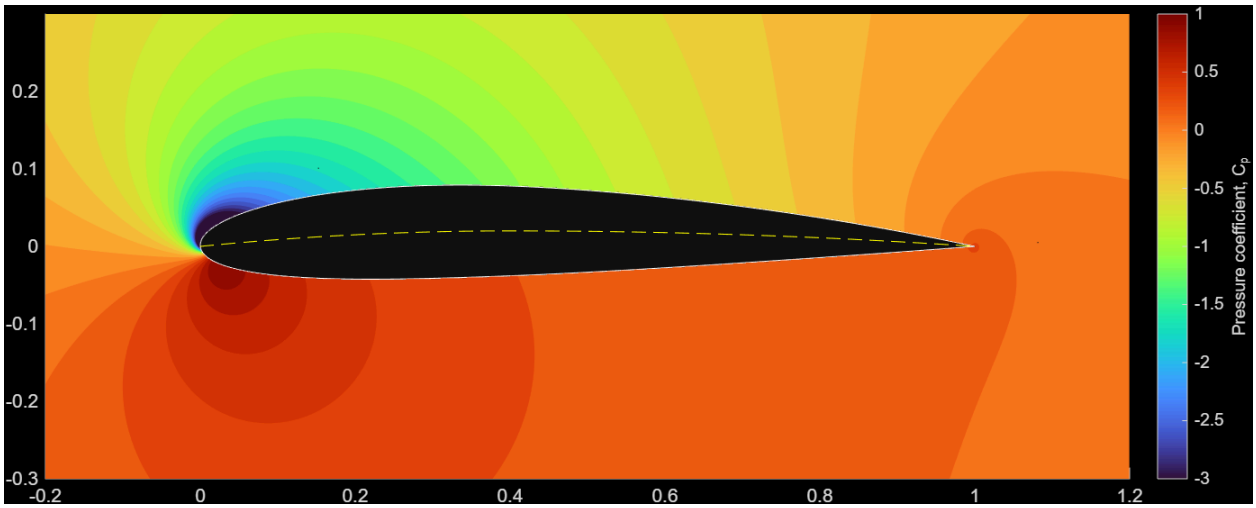
Plot of CP against chord line for different ε. Note that the y-axis is inverted.



We find that on the upper surface aft of x/c = 0.2, all CP predicted from an ε with an exponent lower than -2 converge to the same line. All the CP predictions match on the lower surface at x/c = 0.2.

The greatest difference is in the predictions at the suction peak. This is expected, as there is a steep pressure gradient towards the aerofoil at the suction peak.

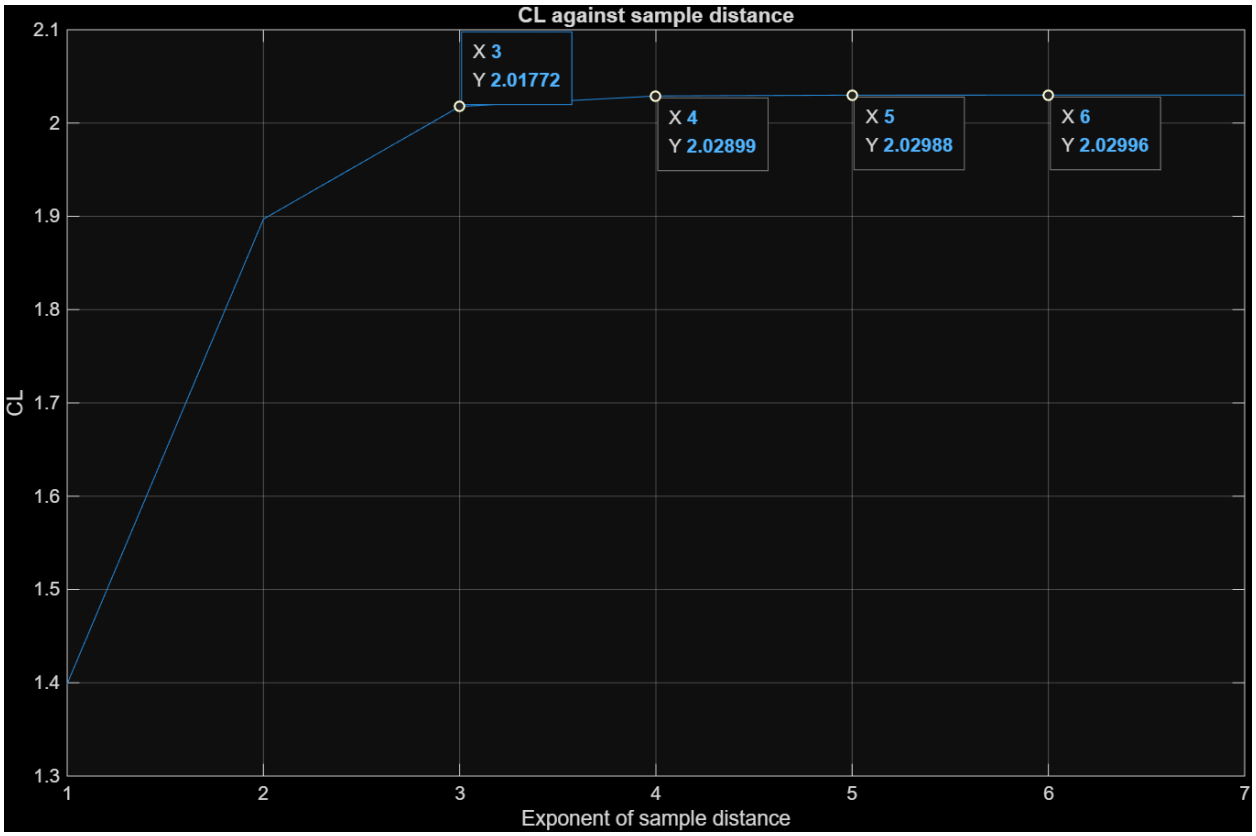
Plot of CP pressure field around a NACA 2412 at 10°



Notice the sharp CP decrease near the aerofoil at the suction peak.

As the pressure in the region of the suction peak is better approximated, the CL computed should also converge to an asymptotic value.

Plot of CL prediction against the exponent of sample distance.



Convergence can be approximated with a power law, CL = CL(∞) + k/ε^q, with exponent q = 0.68. It is obvious that for an exponent of [-1,-2, -3], at a larger sampling distance away from the aerofoil, the CL predicts a value lower than the asymptotic value, mainly due to a lower-than-true suction peak.

Since computation time does not increase as we take a smaller sample (the number of panels, and therefore sample points, is constant), I will use an exponent of -5 for ε in further evaluation. Note that using an exponent of ε too low causes calculations to become singular.

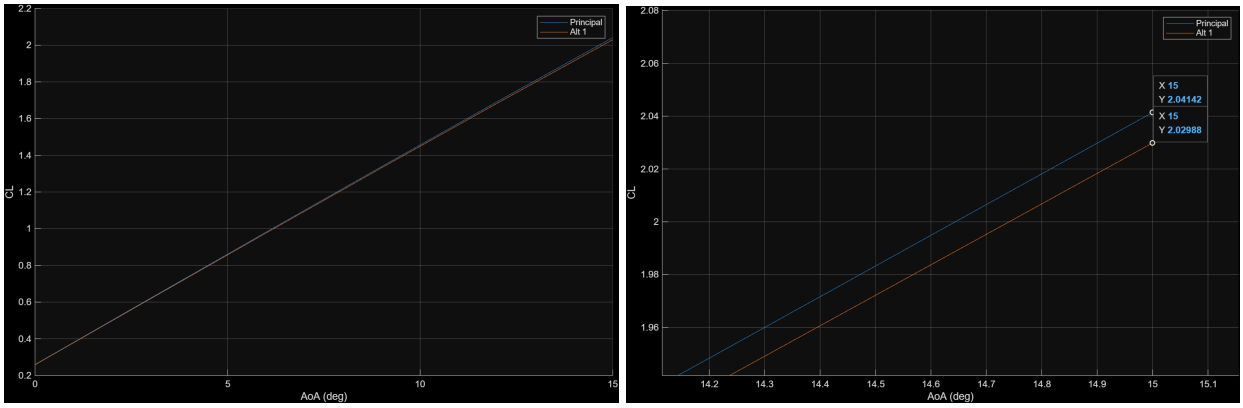
Evaluation of 1st Alternative Method

Method 1 utilises the pressure field around the aerofoil, which allows a pressure plot to be drawn. However, this computation of CL values should not be used over Method 2 due to greater sampling and discretisation errors. Both methods must first calculate the boundary vortex sheet strengths over the aerofoil. Both approximate the vortex sheet strength to vary linearly across each panel. Method 2 then computes the integral of vortex sheet strength over the aerofoil's boundaries as a trapezoidal sum which is faithful to the linear vortex sheet strength approximation.

Method 1 instead calculates the pressure at sample points outside the aerofoil, which is the 1st additional source of error, and further assumes that there is no flow through the control volume sketched by the sample points, a 2nd source of error. Lastly, the integral for pressure force is approximated with a midpoint rule, a 3rd additional source of error.

Note that this method can be considered a simplification of the momentum equation, where the control volume is over the surface of the aerofoil, and the momentum flux is assumed to be 0. Since the control volume is practically over the aerofoil's surface and no-penetration has been enforced, there should be virtually no flow across the control volume's boundaries.

Plots of CL-AoA for Principal Method and 1st Alternative



The prediction of CL from the 1st alternative is lower than the principal method. This could be due to the neglect of momentum flux around the control volume, especially at the trailing edge.

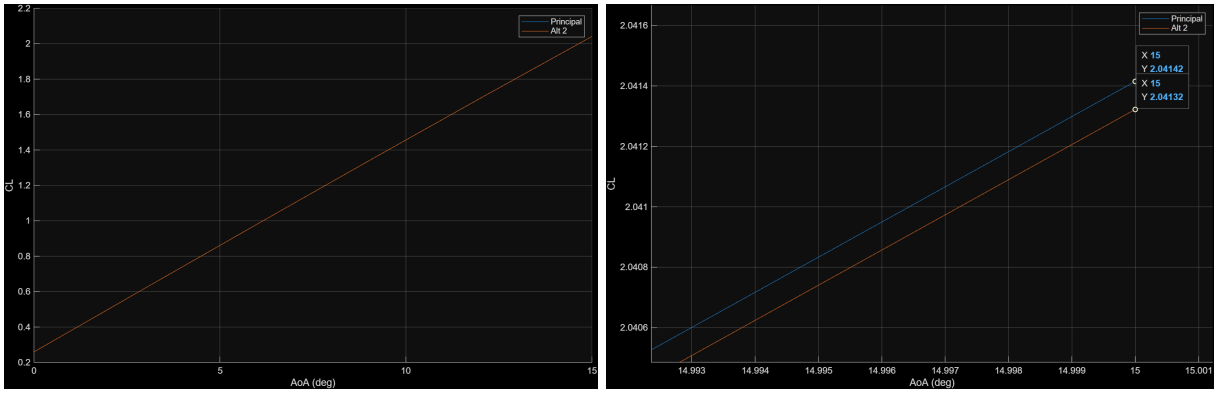
The difference is on the order of 1% and can be considered negligible in comparison with other effects that have been ignored by the Vortex Panel Method.

Evaluation of 2nd Alternative Method

The 2nd alternative utilises the full momentum equation (Body Forces + Surface Forces = Momentum Flux) on a control volume that contains the aerofoil. This serves as a useful test to verify that computations for the velocity field and pressure field are self-consistent. Correct computations should obviously predict a CL equal to the principal method.

A rectangular control volume is used for simplicity. In theory, any control volume that fully contains the aerofoil should yield the same CL. First, a meshgrid is initialised to compute pressure and velocity at nodes. In actuality, only the outer nodes of the grid are required. The pressure over the outer nodes sums to the surface force over the control volume. A trapezoidal sum is used to approximate the integral over the surface. The momentum flux is computed in a similar manner but instead uses a midpoint rule.

Plot of CL-AoA for Principal Method and 2nd Alternative



As shown, the calculated CL is extremely close to that used in the principal method (<1e-2%). It may be theorised that, therefore, the previous alternative's neglect of the momentum flux was a great source of error. A node spacing for the meshgrid of 0.05 was used. Note that if the control volume has boundaries close to the suction peak, errors are increased slightly due to the sudden changes in pressure. However, the difference with the principal method remains less than 0.01%.

An appropriately far control volume is -0.2<x<1.2 and -0.5<y<0.5. Note that increasing the boundaries increases the number of sample points and may increase computational time.

References

https://v0xnihili.github.io/xfoil-docs/formulations/#viscous

Houghton, E.L., Carpenter, P.W., Collicott, S.H. and Valentine, D.T. (2013) Aerodynamics for Engineering Students. 6th edn. Butterworth-Heinemann.